

1. Efficacy by Claim

- **Strength/power (with resistance training): Strong evidence.** Numerous meta-analyses and RCTs show creatine + training boosts strength and power more than training alone. For example, a 2024 meta-analysis (16 RCTs) found ~4–11 kg greater gains in bench press/squat strength with creatine vs. placebo [26†L193-L202]. Another review in young adults reported large strength improvements with creatine [26†L193-L202]. Overall this is a well-replicated finding (multiple meta-analyses, ISSN position stands).
- **Lean muscle mass (with resistance training): Strong.** Meta-analyses consistently find modest but significant extra lean mass gain on creatine. For instance, one review found creatine+training increased lean body mass ≈ 1.1 kg more than training+placebo [33†L95-L103] (95% CI ≈ 0.6 – 1.7 kg). A Bayesian meta of ultrasound muscle thickness found small increases favoring creatine (~ 0.1 standardized units or ~ 0.13 – 0.16 cm thickness) [9†L466-L475] [9†L485-L494]. All high-quality (pooled) data indicate a real hypertrophy benefit with creatine.
- **High-intensity intermittent exercise (sprinting/HIIT): Moderate.** Creatine is mechanistically expected to aid short bursts of power (via phosphocreatine buffering). Many trials report improved power output or repeated-sprint performance, though fewer large meta-analyses focus on these outcomes. For example, some studies show higher Wingate peak/mean power on creatine. The overall verdict is positive but less quantified than for strength: experts note creatine “*augments power output*” and anaerobic performance [112†L1-L4], but published meta-analyses are fewer. We judge the evidence moderate.
- **Recovery from exercise:** Weak/mixed. A meta-analysis found creatine modestly blunts muscle-damage markers (e.g. CK) after an *acute* exercise bout [40†L103-L112]. However, with chronic training the effect disappears or even reverses; creatine users sometimes show higher CK at 24h during long-term training [40†L103-L112] [41†L71-L79]. Importantly, improvements in functional recovery (e.g. return of strength) have **not** been clearly demonstrated [41†L71-L79]. In short, creatine may slightly reduce damage markers acutely, but there is little evidence it speeds real recovery of performance.
- **Cognitive/brain effects:** Moderate/weak. Emerging research suggests creatine can enhance some cognitive tasks, especially under stress. A 2024 meta-analysis (16 RCTs) found significant benefits for memory (standardized SMD ≈ 0.30) and attention/processing speed [45†L413-L422] (moderate-certainty evidence for memory). However, no overall improvement was seen for *general* cognition or executive function [45†L419-L428]. Subgroup analyses hint greater gains in certain groups (e.g. younger adults, some patient groups) [45†L413-L422]. Individual trials show creatine can offset cognitive decline during sleep deprivation [47†L191-L199] or improve memory in vegetarians [48†L311-L315]. In sum, creatine **may** help memory/processing speed in some situations, but evidence is still limited and mixed (more data needed).
- **Other common claims:**
 - **Fat loss:** Insufficient. Creatine does *not* directly burn fat. Meta-analyses find either no change in fat mass or trivial changes. One analysis noted a very small reduction in body-fat percentage ($\sim 0.3\%$) in young adults doing resistance training [55†L12-L20], without any actual fat-mass loss. In practical terms creatine does not cause weight/fat loss – any body-fat differences are likely due to water/muscle effects or chance.

- **“Energy”:** Marketing spin. There is no direct evidence creatine boosts “energy” in the sense of stimulation. It can increase muscle work capacity, but it’s not an energy drink or stimulant.
- **Anti-aging/muscle preservation:** Limited. Creatine can help preserve muscle in aging by aiding strength training, but as an “anti-aging” supplement alone it’s unproven beyond those muscle effects.
- **Mood/mental health:** Preliminary. A few small studies suggest creatine may augment antidepressant therapy or improve depression scores (e.g. creatine + SSRI showed greater mood improvement in women) [56†L1-L4] . These are early findings; more RCTs are needed before concluding a real benefit.

In summary, **strong evidence** supports creatine for strength/power and lean mass with training [26†L193-L202] [33†L95-L103] . Evidence for high-intensity exercise benefits is moderate. Recovery benefits are weak/mixed [40†L103-L112] [41†L71-L79] . Cognitive effects are possible but not definitive [45†L413-L422] [47†L191-L199] . Claims like fat loss are unsupported [55†L12-L20] and others (energy, anti-aging, mood) remain unproven or very preliminary.

2. Effective Dose Ranges by Goal

- **Standard loading:** A common protocol is a *loading phase* of ~20 g/day (often 4×5 g) for 5–7 days [9†L400-L409] . This accelerates muscle saturation. Alternatively, a weight-based loading of ~0.3 g/kg/day (≈ 21 g for a 70 kg person) has been used.
- **Maintenance:** After loading, a maintenance dose of ~3–5 g/day (≈ 0.03 g/kg/day) is typical [76†L59-L67] . This level maintains muscle creatine stores once saturated. Continuous intake of ~3–5 g/day without loading will also eventually saturate muscles (over 3–4 weeks) but more slowly than with loading. Meta-analyses generally use 3–5 g/d as standard.
- **Body-weight/sex differences:** Dosing is often fixed (3–5 g) rather than strictly body-weight-adjusted, but larger individuals might benefit from higher end of range. For example, some protocols use ~0.1–0.14 g/kg/day (≈ 7 –10 g/d for a 70 kg person) as an alternative to 5 g, especially in older adults [76†L59-L67] . Sex per se does not change dosing guidelines. Athletes or heavy lifters may simply use 5 g/d.
- **Goal-specific doses:** The same 3–5 g range is used for strength, power, or muscle gains. Some cognitive studies use higher (e.g. 0.2–0.3 g/kg) for short-term effects. A recent review noted that to elevate brain creatine, ≥ 20 g/day (short term) or ≥ 4 g/day long term may be needed [76†L69-L72] , whereas muscle effects occur with standard doses. In practice, most users take 3–5 g for general performance/health.
- **Minimum effective dose:** Evidence suggests at least ~3 g/day is needed for ergogenic/muscle effects [76†L59-L67] . Doses much lower (e.g. 1–2 g/d) probably have little physiological impact. In other words, a product containing only a gram or two per serving is likely underdosed. The commonly recommended daily amount is 3–5 g, so if a label lists far less (or hides creatine in a blend), the dose is essentially decorative.

Summary: For strength/muscle gains, a typical regimen is a 5–7 day loading (≈ 20 g/day) then 3–5 g/day maintenance [9†L400-L409] [76†L59-L67] . Similar doses apply to most goals. Body-weight scaling (0.1–0.3 g/kg) can be used, but in practice a flat 3–5 g suffices for most adults. Lower doses (<2 g) are unlikely effective, whereas doses up to ~10 g/day (especially with loading) have been studied.

3. Forms and Bioavailability

- **Creatine monohydrate (CM):** The gold-standard form. Well-studied, highly bioavailable, cheap. Virtually all efficacy research uses CM or equivalent. It absorbs well (with ~99% of a dose taken up by muscle over time) and is stable as a powder.
- **Creatine HCl:** Claimed to be more soluble. In practice, studies show 3 g HCl has no performance advantage over an equivalent dose of monohydrate [79†L93-L101] . In one trial, 3 g HCl yielded no greater gains or blood levels than 20 g CM [79†L93-L101] . In short, no human data support superior efficacy. It may mix more easily in water, but absorption and muscle uptake appear similar to CM.
- **Buffered “alkaline” creatine (Kre-Alkalyn):** This is creatine with higher pH. Despite marketing, human studies show no superiority. Claims of no loading needed or less degradation are not backed by independent trials. Researchers generally consider it no better than plain CM, and often more expensive.
- **Creatine ethyl ester:** Deceptive. It was hyped as more absorbable, but evidence shows ethyl ester is actually hydrolyzed to creatinine in the stomach, yielding lower muscle creatine uptake than CM. No credible human study finds ethyl ester superior. It is effectively inferior due to rapid breakdown.
- **Creatine salts (citrate, malate, etc.):** These are creatine bound to organic acids. They are more soluble in water than CM, but doses typically used (e.g. 3–5 g creatine equivalent) mean much of the compound is non-creatine weight. No studies show any performance benefit over CM; most find similar muscle creatine gains or none. Citrate/malate forms cost more per gram of creatine.
- **“Creatine nitrate”, magnesium creatine chelate, other novel forms:** Again, no convincing human data of superiority. They may have slightly different solubilities or taste, but with respect to muscle/strength outcomes, CM remains as or more effective. Until a head-to-head RCT shows clear benefit of a special form, the consensus is that alternative forms are mainly marketing.

Solubility/Stability: Many alternate forms boast higher water solubility, and indeed creatine HCl or nitrate dissolve more readily in a glass than CM. However, solubility alone does not increase muscle uptake. All forms in solution ultimately must cross into muscle similarly once ingested.

Stability (liquid): Creatine in solution slowly converts to creatinine (inactive), especially at low pH or warm temperatures. In powdered form CM is very stable (years shelf life) [80†L9-L13] . In liquids (especially under acidic or warm conditions), some degradation occurs. For example, a study of a milk-based creatine beverage found ~10% creatinine formation at 4 °C (refrigerated) after an initial drop, but continued degradation at room/hot storage [81†L62-L70] . This means creatine-fortified drinks should be refrigerated and consumed relatively fresh. Normal pasteurization (brief high heat) causes only minor loss (<5%) [81†L62-L70] . Fermented products (cheese, yogurt) can contain creatinase enzymes that degrade creatine to creatinine over time [81†L69-L70] , so shelf life is a factor.

Summary: No commercially sold form has proven better bioavailability or efficacy than plain monohydrate. All creatine supplements ultimately supply creatine (or its precursors), and CM has overwhelming evidence. Special forms may mix differently, but none reliably increase muscle creatine beyond what CM does. Creatine in liquid dairy beverages is moderately stable when chilled; significant breakdown can occur with heat or long storage [81†L62-L70] , so formulation must consider pH, temperature, and time.

4. Safety

- **General safety:** Creatine monohydrate has an excellent safety record in healthy people. Dozens of studies (including *long-term* trials up to 5+ years) report no serious adverse effects at typical doses. Reviews state that “no compelling evidence” exists of harm in healthy individuals [92†L319-L327] . High-quality position stands (ISSN, NIH) conclude creatine is *generally safe and well-tolerated* at recommended doses.
- **Tolerable Upper Limit:** No official UL has been set by agencies like NIH or EFSA. This is because studies up to ~30 g/day for years show no clear toxicity in normal users [76†L59-L67] [92†L319-L327] . In practice, keeping to 3–5 g/day (or a short-term 20 g loading) is advised, but no strict upper limit exists since no dose-dependent harm threshold has been identified.
- **Kidney function (“creatine harms kidneys” myth):** In healthy people, abundant research finds creatine does **not** impair kidney function. Meta-analyses show no significant changes in glomerular filtration rate or serum creatinine beyond the expected rise in blood creatinine (from more dietary creatine) [92†L319-L327] . Thus the “kidney damage” concern in normal users is unfounded. People with kidney disease are typically advised to avoid creatine or consult a physician, simply because data in that group are sparse (liver/kidney patients usually take caution with all supplements). But in healthy adults, studies consistently show no adverse renal effects [92†L319-L327] .
- **Other cautions:** Some individuals experience GI upset when taking very high doses (especially when taking >10 g at once); splitting doses or lowering them mitigates this. Unlike many stimulants, creatine does *not* dehydrate or cause cramps – in fact trials often find *fewer* cramp complaints among creatine users [92†L319-L327] . There are no well-known drug interactions (apart from the general advice for kidney patients). As a precaution, creatine is usually *not* recommended in pregnancy or breastfeeding due to lack of safety data, though a few case reports (e.g. in rare metabolic disorders) show no obvious harm [93†L13-L18] . Athletes subject to drug testing should buy third-party certified products (to avoid contamination) but the creatine itself is not banned.

Example consumer statement: “Creatine monohydrate is safe for healthy adults when used at recommended doses. Studies have used up to 5 g/day (or short-term loading doses) for years without evidence of harm [76†L59-L67] [92†L319-L327] . It does not damage the kidneys or liver in healthy people. If you have pre-existing kidney problems or are pregnant, check with a doctor before using creatine. Otherwise, typical use (~3–5 g/day) is well-tolerated, and any occasional stomach upset can be avoided by splitting the dose.”

5. Worldwide Benchmark Products

Below are examples of popular creatine supplements in each region, focusing on pure creatine products (primarily monohydrate). **Note:** third-party certification (NSF Certified for Sport® or Informed-Sport®) is indicated where known.

- **United States:**
- *1st Phorm Micronized Creatine Monohydrate (500 g tub)* – Contains Creapure® monohydrate. 5 g per scoop, 100 scoops. Price ≈ \$0.50–\$0.60/serv (500 g ~\$30–\$60). NSF Certified for Sport® [99†L89-L97] .

- *Athlean-RX Creatine Monohydrate (455 g)* – Unflavored powder, 5.05 g/scoop (~90 servings). Price ~\$0.60–\$0.80/serv (\$30–\$40 per 455 g). NSF Certified for Sport® 【102†L71-L78】 .
- *Momentous SignatureSpec® Creatine Monohydrate (450 g)* – 5 g/scoop, 90 servings. Premium brand often ~\$50 (≈\$0.56/serv). NSF Certified for Sport® 【104†L69-L73】 .
- *Thorne Creatine (208 g)* – Micronized CM, 3 g/scoop, ~60 servings. Market price ~\$20 (≈\$0.33/serv). NSF Certified for Sport®.
- *Transparent Labs Creatine HMB (500 g)* – 3.8 g CM + 1 g HMB per serving (90 serv); price ~\$40–\$50. NSF Certified for Sport®.

• **Canada:**

- *NOW Sports Creatine Monohydrate (500 g)* – Unflavored CM, ~100 servings. ~CAD \$40 (≈USD \$28). (Certified GMP, not specifically NSF).
- *Kaged Creatine HCl (60 g)* – 1.5 g/scoop (20 serv), price ~USD \$35 (\$2/serv). (No known certification).
- *Canadian Protein (Bulk brand) Creatine Monohydrate (300 g)* – 5 g/serv, 60 serv, ~\$30 (USD \$20).
- *UEI Nutrition (Magnum) Creatine Monohydrate (500 g)* – 5 g/serv, NSF Informed-Sport certified, ~100 serv, ~\$35 (USD \$26).

• **Australia:**

- *Bulk Nutrients Pure Creatine Monohydrate (500 g)* – 5 g/serv, 100 serv. 100% Creapure. ~AUD \$40 (USD \$26). No specific sport cert, but third-party tested.
- *Musashi Creatine Monohydrate (300 g)* – 5 g/serv, 60 serv, ~\$40 AUD (USD \$26).
- *KMD Sport Advanced Creatine (250 g)* – 3 g/serv, 80 serv, ~\$30 AUD (USD \$19). (Australian brand).
- *Mr. Vite (Pinnacle) Creatine Monohydrate (500 g)* – 5 g/serv, 100 serv, ~\$50 AUD (USD \$32).

• **European Union:** (major examples)

- *Myprotein Creatine Monohydrate (Creapure®) (500 g)* – 5 g/serv, 100 serv. ~€20 (USD \$22). Third-party tested (Creapure).
- *Optimum Nutrition Micronized Creatine (300 g)* – 5 g/serv, 60 serv, ~€25 (USD \$27). No sport cert.
- *Olimp Creatine Xplode (300 g)* – 3.8 g/serv, 40 serv, ~€20 (USD \$22). Popular in EU.
- *Bee Proud Creapure (Germany) (500 g)* – 5 g/serv, 100 serv, ~€22. (Creapure label ensures purity).

• **Israel:**

- *Imported brands:* Olimp (e.g. Creatine 4000, 300 g) and MyProtein supplements are common (often Creapure). Prices ~NIS 90–130 (USD 25–35) for 300–500 g. Certification varies.
- *Local/retail:* Tnuva's GO protein drinks (mentioned below) are an unusual example. Otherwise, most creatine is sold as imported powder. (Precise Israeli product data was scarce; retailers typically carry the international brands above.)

(Certification note: NSF-Certified for Sport® or Informed-Sport® marks are uncommon outside elite sports products. MyProtein's Creapure is quality-guaranteed but not sport-certified. The above prices are approximate and may vary.)

6. Label-Honesty and Dosing Clarity

- **Effective dose per serving:** As noted, at least ~3 g of creatine per serving is needed for an ergogenic effect [76†L59-L67] . A consumer should look for products listing the actual grams. If a scoop is labeled e.g. 5 g and product is pure creatine, that is adequate. If only 1–2 g is listed, it is likely sub-therapeutic.
- **“Fairy-dusted” products:** The risk is greatest in multi-ingredient blends (e.g. some “pre-workout” mixes). Pure creatine supplements almost always list the full dose. But if creatine appears inside a proprietary “blend” without a breakdown (e.g. “Performance Matrix” of which 5 g total, ingredients not dosed), you cannot verify how much creatine is in there. This is usually a red flag. Unlike caffeine or vitamins, creatine is typically effective only above 3 g; anything less is near useless.
- **Disclosure practices:** Honest labels will show “Creatine Monohydrate — 5 g” per serving, or similar. (Some smaller-serving products use 3 g, which is on the lower end but still justifiable if taken multiple times.) For a blend, you’d need the blend total to be high enough that the hidden creatine could realistically approach 3–5 g, but best practice is that pure creatine products are not “blended” at all. In practice, virtually all top creatine-only products fully disclose the 3–5 g dose; hidden blends with creatine are rare.

Takeaway: The consumer should verify “creatine 3–5 g per serving” on the label. If creatine is in a proprietary blend without a listed amount, assume the dose is not guaranteed.

7. Creatine in Dairy/Protein Drinks

- **Typical doses:** Functional dairy drinks containing creatine generally add only small amounts. For example, Israel's Tnuva GO protein drink (340 mL, 25 g protein) contains 2 g creatine monohydrate per bottle [110†L139-L144] . The label notes a “max daily quantity” of 3 g, suggesting 2 g is the intended serving dose. This is below the ~3–5 g typically recommended daily. International analogues are rare, but when creatine is added to protein beverages (ready-to-drink smoothies, yogurts, etc.), doses are often on the order of 1–3 g per serving – generally insufficient to reach the effective range unless multiple servings are consumed. In short, most creatine-fortified dairy products use “token” amounts (1–2 g/serving), far lower than the 3–5 g daily effective range identified earlier [76†L59-L67] [110†L139-L144] .
- **Dose adequacy:** Because the typical added dose is <3 g, these products alone do not meet an efficacious dose. A person drinking one creatine-enriched dairy beverage per day would likely need additional creatine to reach the effective 3–5 g. It appears the industry currently uses low doses (perhaps for labeling or taste reasons), so the amounts in such products are sub-clinical.
- **Formulation issues:** Creatine is generally compatible with dairy proteins and sugars. No specific antagonism is known – adding creatine to milk or yogurt mixes poses no known absorption issue. Standard pasteurization or UHT heating does not destroy creatine (Uzzan et al. found <5% loss even at 138 °C) [81†L62-L70] . However, storage is a concern. As noted above, creatine can slowly degrade to creatinine in liquid, especially at warm temperatures. In milk drinks, keeping the product refrigerated will largely preserve creatine (only a minor drop) [81†L62-L70] . If kept warm or on

shelf long-term, degradation accelerates (Figure shows ~first-order loss at 23–35 °C) 【81†L62-L70】 . There is also some evidence that *fermentation* (e.g. in yogurt or cheese) can lead to creatinase enzyme activity that degrades creatine 【81†L69-L70】 . Manufacturers of creatine drinks must therefore consider packaging (opaque vs transparent), pH, and storage temperature to minimize creatine loss.

- **Other interactions:** There is no known negative interaction between creatine and dairy protein or carbohydrates; the main issue is stability. Creatine's solubility is limited, but in a formulated drink it is usually in powdered form until reconstitution. Overall, formulating a protein-fortified dairy beverage with creatine is feasible, but preserving full potency requires attention to shelf conditions.

Summary: Dairy-based protein drinks with added creatine typically contain only ~1–2 g per serving 【110†L139-L144】 , below the commonly effective range. Thus they serve more as “for fans” ingredients rather than delivering a full dose. Creatine in such beverages is heat-stable under normal pasteurization, but can degrade over weeks at warm temperatures 【81†L62-L70】 . Refrigeration and appropriate pH buffering help maintain its stability.

Sources: All factual claims above are supported by peer-reviewed research, reviews, or authoritative bodies 【26†L193-L202】 【33†L95-L103】 【40†L103-L112】 【45†L413-L422】 【76†L59-L67】 【81†L62-L70】 【92†L319-L327】 【110†L139-L144】 . Where evidence is mixed or limited, this is explicitly noted.
